

ADE Tokenomics

Input · Output · Context · Usage Strategies

Not a pricing paper. Dollar metering is SpendGuard — see companions.

This paper: how tokens *move* through the harness — what to send, what to emit, what to keep in the window, when to compact / hand off / escalate.

Edition: 2026-07-23 · **Schema:** ade.tokenomics/v1

Inputs: TOKEN_ECONOMICS_RESEARCH.md · Effort B0–B4 · T0–T3 assembly · Continuity · Anthropic context eng / caching docs · Chroma context rot

Deep dive companion: [Token Usage, Compaction & Fertility](#) — languages, formats, custom-language limits, compaction science

0. Vocabulary (lock this)

Term	Means	Does <i>not</i> mean
Tokenomics	Strategies for input composition , output discipline , context window hygiene , and usage patterns across turns	Invoice math, USD rates, SpendGuard honesty
Effort budget	Per-turn gas: tool rounds + output tokens	"How smart the model is" or \$ caps
SpendGuard	Session/daily \$ hard stops; reserve → reconcile	Context compaction policy
Context	What is in the model window <i>this round</i> (system + history + tools)	Disk state / .ade/ identity
Cache-stable prefix	Ordered static bytes that prompt-caching can hit	"Any long prompt"
Fertility	Same text → fewer/more tokens across tokenizers/models	Quality by itself (EMNLP 2024: compression ≠ accuracy)

Orthogonal tanks (always):

tool rounds ≠ output tokens ≠ context occupancy ≠ \$ spend

Wrong failure mode = wrong fix. Empty Effort → Continue / thrift. Full window → compact / Continuity. Empty \$ → SpendGuard halt. Don't swap models to "fix" a context problem.

1. Thesis

Tokens are the **discrete channel** between host and model. ADE cannot rewrite a hosted tokenizer. It *can* decide:

1. **What enters** the channel (rules, skills, files, history, tool junk).
2. **What leaves** (verbosity, diffs vs essays, tool spam).
3. **What persists** across rounds (growing transcript vs capsule).
4. **When the loop thrifts** (host next_safe_command, handoff) vs pays for rediscovery.

Quality ≈ model × harness × human process still holds. Tokenomics is the harness's **context & I/O discipline** — the middle term applied to the channel.

2. The agent-loop shape (why strategy matters)

Each tool round typically **re-sends** a growing prefix (system + history + prior tool results). Mechanism:

cost_shape ≈ 0(rounds × growing_context) // not 0(rounds)

So:

- Dumping "helpful" context early **taxes every later round**.
- Unbounded tool-result accumulation is **quadratic in practice**.
- Blank regenerate / "continue" without a next step buys **another full rebill** of the same rot.

ESTIMATE (workload-dependent): multi-round coding agents often use **3–8×** tokens vs a single-shot call before useful completion. Measure on ADE gold / dogfood — don't treat as law.

Cached tokens still occupy the window. Caching changes **provider price**, not capacity. Tokenomics cares about **occupancy and recall**; SpendGuard cares about **invoice class**.

3. Input strategies (what to send)

3.1 Progressive disclosure (HIGH pattern)

Layer	Content	When
T0	Tiny start contract (~300–400 tok); points at activate_skill	Always
Always-on rules	Deny/invariants only (secrets, money, build-risk)	Always, soft-budget truncated
Catalog	Skill names + one-liners	Always (cheap index)
T2 / match	Keyword / always_apply skill bodies	Only when matched — keep thin
T3 activate	Full skill + capped references/	On tool call mid-turn
Handoff	Capsule summary, not full chat paste	On Continuity resume

Anti-pattern: stuffing every procedure into always-on rules. That is **context dump with better branding**.

3.2 Cache-stable ordering (HIGH — provider docs)

Put **static** content first; **dynamic** last:

```
system / T0
  → tool schemas (stable)
  → AGENTS.md + scoped rules (stable-ish)
  → skill catalog (stable)
    → eng-goal / owned paths
    → handoff capsule
    → user turn
    → tool results (most volatile)
```

Breaks the prefix (avoid early): timestamps, random IDs, per-turn “now”, churning model ids, unsorted file dumps, “session UUID” in system text.

Minimum cacheable prefix is commonly ~**1,024** tokens (provider-dependent). Write tax (cache write > base input) means **low hit rate can cost more** than no cache — measure hits.

3.3 Retrieval beats dump (HIGH)

Instead of	Do
Paste whole module	Search → symbol → short excerpt
“Here’s the repo”	Scoped paths + lease owned_paths
Re-paste last 40 turns	Continuity capsule + thrift resume
Always-on 8k skill bible	Catalog + activate_skill

3.4 ADE soft input budgets (shipped)

From context.rs defaults (chars÷4 **heuristic only** — never bill on this):

Bucket	Soft tokens
Always-on	400
Rules	2,400
Skills	6,000
Handoff	500
Status	Green / Warn@85% / Critical

Gap: soft truncation of **system assembly** ≠ mid-thread **history** compaction. History still grows unless Continuity / human discipline intervenes.

4. Output strategies (what to emit)

4.1 Constrain outputs first (HIGH)

On frontier coding APIs, **output is typically ~4–6× input** per token. Thinking/extended reasoning is usually billed as output-class. Therefore:

- 1. Cap **max output / Effort dial** before micro-optimizing a 200-token system trim.
- 2. Prefer **structured short acts** (diff, tool call, next_actions chips) over essays.
- 3. Verifier slots should emit **pass/fail + evidence pointers**, not regenerated patches.

4.2 ADE Effort = output + rounds gas (shipped)

Effort	Tool rounds (typical)	Output cap (typical)
Low	16	~4k
Med	24	~16k
High	32	unlimited (provider limit)

Implementation intent (session.rs): Effort counts **generated (output) tokens only**. Counting input each round would re-bill the system prompt and kill Apply mid-task — that is correct **tokenomics**, not a loophole.

Apply / Automate / Continuity resume floor **Med+** so thrift resumes aren’t starved.

4.3 Output anti-patterns

Anti-pattern	Better
Narrate every tool	Tool call + one-line why
Paste full file “for clarity”	Patch / hunk only
Self-certify in prose	Run verify; quote sensor
“Here’s everything I considered”	Decision + next_actions ≤3
Regenerate whole answer on typo	Targeted fix turn

5. Context window strategies (what to keep)

5.1 Context rot (HIGH — Anthropic / Chroma line of work)

Larger windows ≠ reliable use of mid-context. Models **under-attend** buried instructions and facts as occupancy grows. “Just use 200k” is not a strategy.

Tokenomics response: keep the **working set small and fresh**; move durable state to disk (.ade/goals, leases, capsules), not to chat immortality.

5.2 Compaction policy (ideal)

Trigger	Action
~ 70% window (or Warn→Critical trend)	Compact: drop stale tool blobs; keep decisions, paths, failing tests, open questions
Scope change	New thread + capsule link — don’t stretch one transcript across two jobs
Budget exhaust (rounds/out)	Continuity capsule; host may run next_safe_command; thrift resume
Dual concern (explore + mutate)	Split Scout vs Worker sessions — don’t mix discovery dumps into Apply history

ADE today: Continuity + compaction **metrics** on capsules; **not** automatic mid-thread history compaction. That is a first-class Tokenomics gap.

5.3 What belongs on disk vs in-window

On disk (.ade/)	In-window
Goals, leases, tasks, rules, skills bodies	Active goal summary
Capsules / chat thread archive	Last decisions + open blockers
Verify evidence pointers	Failing command + excerpt
Spend ledger	“Approaching cap” chip only

6. Usage strategies (how humans + harness spend turns)

6.1 Good turn shapes

Shape	When	Tokenomics effect
Clarify ≤3	Underspec before Apply	Avoids expensive wrong mutate
Suggest → Apply claim-one	Multi-step work	Planner history ≠ worker history
Verify gate	Automate / done claims	Sensors replace essay self-grades
Continue + thrift	Effort empty mid-mission	Pays for <i>progress</i> , not rediscovery
Isolate worktree	Parallel / risky Apply	FS isolation; don’t duplicate context dumps

6.2 Bad usage (human × loop)

Habit	Channel effect
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Blank "try again"	Full rebill; same rot
Context dump "to be safe"	Permanent tax every round
Mid-session model swap	Cache miss + distribution shift
New chat, same vague task	Re-pays exploration
Continue with no next step	Pure overhead turn
Paste prior chat into Continuity	Defeats thrift; rebuilds dump

6.3 Multi-agent tokenomics

Peer agents that **restate** each other’s context multiply occupancy without multiplying truth (CooperBench coordination tax is quality; tokenomics adds **N× rebilled prefixes**).

ADE contract: Planner ≠ Worker ≠ Verifier; **claim-one**; leases. Share **structured task + capsule**, not full peer transcripts.

7. Fertility & model choice (same text, different tokens)

Tokenizers differ. Newer families can emit ~+30% tokens for identical text (vendor notes — verify per card). **Fertility** = tokens per unit meaning on *your* corpus.

Do	Don’t
Pick models with fertility + quality on ADE gold	Assume fewer tokens ⇒ better
Lock model for a session’s cache prefix	Silent swap mid-task
Route Scout to small / local	Use frontier for every 1s

EMNLP 2024 reminder: tokenization-as-compression does **not** reliably predict accuracy. Fertility is an **I/O efficiency** knob inside a quality floor — not a free lunch.

8. Playbook — best ways (ranked for ADE)

Host-enforceable first; prompt pleading last.

Rank	Practice	Why
1	Thin always-on + activate skills	Stops permanent input tax
2	Cache-stable prompt order	Hits when providers cache; keeps prefix sane even without \$
3	Orthogonal Effort (rounds + out-tokens)	Right stop; Continuity path
4	Continuity thrift + host next_safe	Progress outside rediscovery loops
5	Contract / clarify before Apply	Wrong work is the worst token spend
6	Retrieval over dump	Protects every future round
7	Output discipline (short acts, verify not essays)	Output is the expensive side of the channel
8	Compact @~70% / scope-split threads	Fights rot and O(N×rounds)
9	Model lock + profiled routing	Cache + role-fit

10	Provider usage for truth; chars÷4 for assembly only	Don't optimize a fake meter
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SpendGuard sits beside this list for \$ — it does not replace any row above.

9. ADE scorecard — shipped vs Tokenomics gaps

Capability	Today	Gap
T0–T3 + activate_skill	Shipped	Tighten always_apply / keyword bloat
Soft system budgets + Warn/Critical	Shipped	History compaction @~70%
Effort rounds + output caps + honesty	B0–B4 done	—
Continuity + host thrift	Shipped	Default auto-compact trigger
Cache-stable order	Documented	Enforce + cache-hit telemetry
chars÷4 assembly	Shipped	Label in UI; never bill
Fertility bench	Planned	Gold fertility suite
Model profiles / lock	Stub / manual picker	ade.model-profile/v1
Output style policy	Effort maxTokens	Stronger “short act” host norms
\$ honesty	Separate track (H1)	Not Tokenomics core

10. Design rules (normative for ADE)

1. **Tokenomics ≠ Spend.** Never name a \$ feature “tokenomics.”
2. **Disk is cheap; window is sacred.** Persist goals/leases/evidence; keep working set small.
3. **Static first, volatile last.** Protect the prefix.
4. **Catalog > body.** Bodies on activate.
5. **Output before input micro-trims.** Cap essays and thinking.
6. **Count output for Effort; count provider usage for \$.**
7. **Empty tank → Continuity, not Provider error, not silent model swap.**
8. **Compact or split before rot.** ~70% or scope change.
9. **Planner history ≠ Worker history.**
10. **Measure on gold / dogfood** before claiming multipliers.

11. Relation to other papers

Document	Role
This paper	I/O · context · usage strategies
TOKEN_ECONOMICS_RESEARCH.md	Internal science notes + \$ adjacency
<i>Agents, Harnesses & ...</i> brief	Harness / MAS / verify; \$ pricing as <i>adjacent</i>
<i>Ideal ADE / What ADE Could Do</i>	Product architecture & bets
<i>Agents Meet Humans</i>	Agnostic human habits (incl. dump / regen)
Effort / SpendGuard plans	Orthogonal tanks

12. One-page cheat sheet

INPUT thin rules · catalog · activate · retrieve · static→dynamic
OUTPUT short acts · Effort caps · verify>prose · no dump-on-regen
CONTEXT working set small · disk for durable · compact@~70% · split scope
USAGE clarify→Suggest→Apply claim-one → verify → Continuity thrift
NEVER mid-task model roulette · paste thrash · bill on chars÷4
ORTHOGONAL rounds ≠ out-tokens ≠ window ≠ \$

End of ADE Tokenomics · ade.tokenomics/v1 · 2026-07-23